Audio file

chapter 24 voiceover full.mp4

Transcript

00:00:01

So this chapter, we are going to go over chapter 24, which is our assessment of our musculoskeletal system.

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So there's going to be a couple of objectives that we are going to go over when we are talking about this chapter.

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Some of the objectives that we will kind of go over is what is the structure and the function of our musculoskeletal system?

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What are the differences between our normal and abnormal findings of our musculoskeletal system?

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And also analyzing interview and physical assessment data that is related to our musculoskeletal system to make our clinical judgment.

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So as we think about our body systems and as we learn about our normal and abnormal findings, sometimes it's important for us to know and understand

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What is the structure and the function, and what is the reason for the function of our bones?

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So our bones in our musculoskeletal system, our musculoskeletal system consists of bones, muscles, and joints.

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And they are controlled by our nervous system, which helps us with movement.

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Our main purpose is to provide structure and movement for our body parts.

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So flexion, extension, adduction, abduction, you guys will learn that in the slides coming forward.

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But our bones, they provide structure and they provide our protection.

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So it prevents us from, it gives us that structure of our body and it consists of 206 skeletal bones.

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It stores calcium.

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So you guys will learn and you will understand there are many assessments to determine if we have enough calcium in our body.

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We can do bone density screenings to make sure that the density of the bones are strong.

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So as you know, our patients get older, sometimes bone density decreases and we're more prone to getting arthritis and osteoarthritis and all of these things.

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So it helps us store calcium.

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and also produces blood cells.

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There are two types of bones.

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There's compact bone and spongy bone.

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So your compact bone is your hard and dense type of bone.

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It makes up the shaft and outer layers.

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And you have your spongy bone, which I like to say is the internal part of the bone.

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And it contains numerous spaces and makes up the ends and the centers of the bones.

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So our bones, there's compact bone and there's spongy bone, helps us protect our structure and protection, serves as levers, stores calcium, and produces our blood cells.

00:03:04

So I mentioned in the previous slides that our musculoskeletal system consists of bones, muscles, and joints.

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This slide kind of talks about what is the structure and the function of our muscles.

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There are three types of muscles in our body, skeletal, smooth, and cardiac.

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This chapter mainly focuses on skeletal type muscles, so the muscles that are around our skeletal bones and kind of cover them.

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The musculoskeletal system overall with our skeletal muscle consists of 650 voluntary voluntary muscles which are

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controlled, conscious controlled by us, by our nervous system.

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The skeletal muscles attach to bones by way of strong fibrous cords called tendons.

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So tendons are that strong fibrous cords that helps attach to the skeletal bone.

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So when you guys are inserting IVs,

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or placing lines in arteries and veins, we have to make sure that we are not hitting a tendon.

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So that is why when we teach you guys skills and we explain the reasoning for the way we look for sites, we have to ensure that we are not feeling a tendon.

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So sometimes a tendon may feel firm and a little bit more hard than what a vein would feel like.

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Our skeletal muscles also help with posture, helps us produce body heat, and allows our body to have range of motion or movement.

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There are a couple movements that our skeletal muscles can help with, and we'll talk about that more in the previous slides.

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There's abduction, there's adduction,

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There's also circumduction, inversion, eversion, extension, hyperextension, flexion, dorsiflexion, plantar flexion, pronation, supination, protraction, retraction, and rotation.

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Those are some of the movements that we are able to do with our body with our skeletal muscles.

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You guys will learn that more in detail in the upcoming slides.

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So the third component of our musculoskeletal system is our joint.

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So I mentioned there's muscle, there's joints, and there's bones.

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So this part is we're going to talk about joints.

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Joints are something we call or articulation is where two or more bones meet.

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The joints helps us provide a variety of different types of range of motion.

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kind of helps us move.

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It's classified in three different types, fibrous, cartilaginous, and synovial.

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So your cartilaginous joints are generally between your vertebrae, and they're joined by different types of cartilage.

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So your synovial joints may be in your shoulders, your wrist, hips, knees, ankles.

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I like to say your synovial joints are filled with fluid.

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So there's some sort of fluid that helps as a lubricant to promote those types of sliding types of movements.

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So for example, if you're rotating the wrists, right, or you're making, having your patient lift their leg up and lift their leg back down, okay, and the ankles, okay, think about what is that synovial fluid doing to help with that range of motion?

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So in addition, your synovial joints are joined by different types of ligaments.

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And ligaments are strong, dense bands of different types of fibrous connective tissue.

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And that kind of helps them kind of come together.

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So remember your structure and function of your joints.

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Joints are where two of your bones meet.

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They kind of help with range of motion.

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So you guys can see in this photo some of the different types of skeletal muscle movements.

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So how our skeletal muscles move.

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Remember I talked about there's abduction, adduction, circumduction, inversion, eversion, extension, hyperextension, flexion, dorsiflexion, plantar flexion, pronation.

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supination, protraction, retraction, and rotation.

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So some of these are shown.

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So rotation.

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Rotation is kind of having the patient move their head back and forth, one side to the left, one side to the right.

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How well can they rotate?

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Then you can see in that second photo, protraction and retraction.

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So you can see that protraction in that second photo, that person in the photo is kind of moving their chin forward.

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So that's protraction.

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Retraction is the ability to move backward with that chin.

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Pronation.

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Think about prone and supine.

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I like to remember prone and supine when I'm remembering these types of muscle movements.

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Okay, so when I think about pronation, pronation is turning kind of your hand or your hand forward.

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So it's turning or facing downward.

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So pronation, your hand is facing downward.

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Supination, your hand is turning or facing upward.

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So if you guys remember pronation, okay, and supination, I like to think of it in positions.

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So prone position, sometimes your patient can be kind of laying on their tummy, so they're facing downward.

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Supine, your patient is facing upward, so they might be laying down with kind of the front of their body facing upward.

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Okay, that's pronation.

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Supination, I talked about protraction and retraction and rotation.

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Next thing we're going to talk about is eversion and inversion.

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So eversion is moving outward.

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So you're moving something outward.

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So you guys see that example?

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It's kind of the ankle with a foot.

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So you're kind of moving it outward.

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Inversion, you're moving it inward, so you're going towards the body.

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Extension and flexion.

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Extension and flexion is something you guys will always assess, especially with your patients with a range of motion.

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So extension, when you guys practice extension, you're going to educate the client or ask the client to straighten their extremity of the joint, and you increase the angle of the joint.

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So when we're thinking about our extension,

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We're straightening out the arm.

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Straighten out the arm at increasing the angle, you're gonna straighten it out.

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That's your extension.

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Your flexion, I like to think of flexing your muscles, right?

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Your flexion is your bending the extremity at the joint, decreasing the angle of the joint.

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So remember, extension, you're straightening the angle

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you're straightening the extremity of the joint, increasing the angle of the joint.

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So you're stretching it out, flexion, you're kind of flexing like flexing a muscle.

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Okay, so with your flexion, you're bending the extremity to decrease the angle of the joint.

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Then you guys see abduction and adduction.

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Abduction is you're moving away from the midline of the body.

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Okay.

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So say for example, you are assessing range of motion of the arm and you ask the patient to raise their arm up.

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Okay.

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That is abduction because you're moving away from the midline of the body.

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Okay.

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Adduction, you're moving towards the midline or adduction.

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You're moving towards the midline of the body.

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So you're going to ask them to raise their arm down close to kind of their hips down to their

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Knees, that is your adduction.

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So remember, abduction, you're moving away from the midline.

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Adduction, you're moving towards the midline of the body.

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You guys can see elevation and depression as well over here.

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Elevation, you can have them raise their shoulder up.

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So you're kind of going in upward position.

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And then you have depression, which you can have the client raise their shoulder down.

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What I forgot to mention as well with that is hyperextension.

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So hyperextension is when the joint can bend greater than 180 degrees.

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The last thing you guys see here is circumduction.

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So circumduction is just a circular motion.

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So you can see in this photo right here that person is kind of going in a circular motion.

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with their hand.

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So for example, ask the client, can you move your arm in a circle for me?

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And what you're assessing is a range of motion in a circular motion.

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So I kind of mentioned abduction, adduction, circumduction, inversion, eversion, extension.

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hyperextension, flexion, pronation, supination, protraction, retraction, rotation.

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So these are some of the different types of movement that our skeletal muscles can have.

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So now we're thinking about our collection of subjective data.

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We're always going to think about what type of data are we collecting to figure out our clinical judgment.

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So when we're assessing our musculoskeletal system, think about the structure and function, right?

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Helps us with range of motion, helps us with movement.

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So think about some of the questions you might ask for patients with range of motion issues.

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What are their levels of functioning, whether activities of daily living?

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So what I mean by activities of daily living is are they able to wash their dishes?

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Are they able to button up their sweater?

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Are clients able to reach up to the counter or reach up to the drawer in the kitchen and pull out a plate?

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Those are activities of daily living.

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Because if we're having issues or complications and we're not, or your patients are not able to do these activities,

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It can influence a lot of the care that we might be giving for these clients.

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Some of the questions you might ask, are we having any pain?

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We're asking pain because think about your arthritis.

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Think about your osteoarthritis.

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Think about those types of disorders that can affect our movement.

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Sometimes it can be painful for them to move a joint.

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or to exercise.

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Any difficulty performing normal activities of daily living.

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So I mentioned some of those already.

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What medications are they taking?

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Are they taking pain medications?

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Because they can't move their joint like they were able to.

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Do they smoke?

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Do they drink alcohol?

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What is their exercise?

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routine.

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Do they walk?

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Do they walk in their neighborhood?

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What do they do?

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Also, we think about our aging adult.

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Because when we ask these things, smoking, alcohol, things like that, why do we ask these things?

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Um, intake of alcohol and increased caffeine can also can increase a lot of the risks of osteoporosis.

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Okay.

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And, um, we're asking our exercise because think about our clients that may not be able to, um, you know, maybe live a very, um, very lazy lifestyle.

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They don't exercise regularly.

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Think about how their bones or their joints or their muscle movements can deteriorate faster than maybe someone that has regular exercise.

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Regular exercise promotes flexibility, helps us with our muscle tone, helps us build that muscle tone and strength.

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So our aging adult, it is really important to ask these questions

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not just for older adults, but for anybody that you may be assessing with any sort of musculoskeletal issues.

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So our aging adult.

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So a lot of our older adults have an increased risk of having decreased bone density.

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So what I'm meaning is your osteoporosis.

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So osteoporosis, our bones start to demineralize

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They can become more porous and fragile, which means that our bones aren't going to be as dense and hard and firm.

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What happens is that bone loss can start occurring and it increases them for fractures, increases their risk for fall, increases a lot of our safety issues with our older adults.

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We're asking, is there any sort of weakness?

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Is there any sort of weakness?

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So meaning are they having difficulty getting around at home?

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Have they had a fall in the last two months in their house?

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Are they having any joint stiffness in their joints causing or related to arthritis?

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Do they use any mobility aids such as a walker?

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right?

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Do they have grab bars at home?

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So these are the type of things we're thinking about when we're asking our older adults, your big safety risk for them.

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They if they're at risk for weakness, their risk for falling, right?

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If their bones are starting to be more fragile, not as strong because their bone density decreases, they're at risk for fractures, right?

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So think about your patients that have

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hip replacements, knee replacements, right?

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Over time those bones and those joints start to deteriorate and we have to think about what are the factors that can cause a lot of these issues.

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So now we're collecting our objective data.

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So what are the findings that we are looking for as nurses?

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So the physical assessment of our musculoskeletal provides what is the client's posture?

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What is the client's gait?

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What is their bone structure?

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How strong are their muscles?

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How well is the patient's ability to move their joints?

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As well as to perform their activities of daily living like you mentioned before.

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So the main purpose of our collection of data is to provide or to assess what is the function for activities of daily living and what are the abnormalities that we have to screen for.

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So when you guys are assessing older adult clients, older adult clients have slower movements.

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They move a lot slower than us young people.

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They have reduced flexibility, so they might not be able to flex or extend as far as other people.

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They can also have decreased muscle strength.

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because of age-related issues with their muscle fibers and joint degeneration.

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They also have reduced elasticity of the tendons and they have more calcification in their joints.

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So what I mean by that is their joints become more calcified.

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So it becomes a little bit more hard, becomes more firm.

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It's harder to move some of those joints.

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So again, assess your posture.

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Assess your gait.

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What is the bone structure?

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So bone structure or do they have small arms, right?

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Do they have any deformities?

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What is your muscle strength, right?

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Weight bearing exercises.

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Are they able to lift a spoon?

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Are they able to do their daily activities?

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What is the mobility of their joints?

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Okay.

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This will give you a lot of data regarding the ADLs through their medical history.

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So look, do they have a history of arthritis?

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Do they have a history of osteoporosis?

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Do they have a history of osteoarthritis?

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So also we have to think about those age-specific screening measures such as risk for falls.

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Older adults as they progress in their age are at increased risk for falls.

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And they're even higher risk for falls, especially if they have different joint disorders, causing them to be more fragile and increasing their risk for any sort of safety risk of falling, injuries, trauma.

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Okay.

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So now we're looking at our collection of our objective data with our inspection and palpation.

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So remember when we're inspecting, we're looking at

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the joint, we're looking at our sights, our sounds, our smells, our surroundings, right?

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You're looking at the skin, you're looking at the joint.

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So look at the feel, look for the size, right?

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Are the bones or are the bones skinny and frail?

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Right?

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So you'll see those little old ladies that are really tiny and they have frail bones and their arms are small.

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What is the contour of the joint?

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What does it look like?

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Right.

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Do they have muscles?

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What does it look like?

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Are they hexic?

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Is that is there not enough?

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Are they not eating enough and they just look very, very thin?

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What is your skin color?

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Okay, what are the characteristics we're looking for?

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Is it appropriate for ethnicity?

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Is there any swelling of the joints?

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Okay, so sometimes if

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Patients are having knee issues or hip issues.

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Sometimes they may have swelling at the site.

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Do they have any masses?

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Right.

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That's your inspection.

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Okay.

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Your palpation, your palpation.

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Remember the technique is you're using two or three fingers.

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You're lightly palpating those areas.

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You're assessing for tenderness.

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Okay.

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So if you feel that site palpate, is there any tenderness?

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Ask the patient, are you having any pain?

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Is there any tenderness that elicits pain when I press on it or when I press on the site that you're assessing?

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Is there any swelling?

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You can also feel the palpation heat.

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So if there's any sort of swelling of the joints or if there is any sort of pain, there could be heat associated with it.

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So if you feel it and you're like, oh, that's a little bit warmer, right?

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You can feel temperature, okay?

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Also, when you're collecting your data, so you have your inspection, your palpation, and this part called range of motion, you guys will always hear this, and you will have to say when you complete head to toe, I'm going to have a couple of range of motion activities for you to perform if you can.

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What I like to say with range of motion exercises when you are completing these with your client is do not push them to the limit.

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Perform only what they can do with their strength.

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Do not try to push them.

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Do not try to keep overextending or overflexing if you're doing different types of range of motion.

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Just really be patient with them and say, do the best you can.

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There's different types of range of motion.

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There's active and there's passive.

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Different types of range of motion are the ones I mentioned in the previous slide.

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So when we're thinking about our range of motion, we can also assess it with that meter that you guys see on the left hand side.

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That is a way that you can assess for different types of rotations or different types of degrees that joints may be able to move.

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Slow muscle strength.

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So every time we do or collect

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different types of our objective data, we have to figure out how do we grade these.

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Do you guys remember when we're grading vital signs and grading pulses, we do the same thing for muscle strength.

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It's graded on a zero to five scale.

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So zero, that patient might be paralyzed and you might do a range of motion with them in the ICU and you lift their arm up and the arm goes right back down.

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Alright, no contraction of the muscles noted and there's no movement.

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Two, okay, one, sorry, let's go back to one.

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That could be your weakness.

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So your patients could be extremely weak.

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So if you try to help with passive range of motion,

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Or you might have to help them and guide them.

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They might be really weak and they might just drop their arms slowly.

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So maybe there's a little bit of a movement or a contraction, but they're still pretty weak.

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It might hurt.

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And they're like, oh, I don't want to keep doing that again.

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Two, that's poor range of motion.

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So you might have to help these patients, right?

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So help them with some passive range of motion where you're helping guide as well.

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So kind of help with assistance.

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So passive range of motion, you remove gravity and they have kind of that poor range of motion becomes pretty weak.

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So you can help them as well.

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Three is your kind of average type of weakness.

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So it says on your slide active movement against gravity.

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So again, that could be you're having them, you know, maybe flex their arm up and then all of a sudden just starts gradually going down, right?

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That's an average amount of weakness that they may have.

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And then four is active motion against some resistance.

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Okay.

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So what I like to kind of assess with when they have some sort of slight weakness,

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okay with some resistance is maybe if I am doing a stroke assessment and I have I asked the client to raise their foot up towards the sky and I kind of put a little bit of resistance on my hand and I tell them try to push up against my hand and I'm assessing do they have any active motion against resistance so are they trying you can feel that

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Is there any resistance?

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Are they giving you resistance or is there no resistance at all?

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Do they have any slight weakness?

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And then five is your normal.

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They have active motion against full resistance.

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So you guys could do this in an arm.

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You guys could do this on a leg.

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You could do it kind of on the feet.

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You guys will learn about that on head to toe.

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So in the arm, you can do the same thing.

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Have the patient raise their arm up.

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And then you can kind of hold your hand down.

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I like to say with a palm towards the hand.

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And I kind of have my light palpation on the hand.

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And I ask my patient when I'm assessing number five,

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Okay.

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Try to push up against my hand and try to resist as I'm pushing down on your arm.

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Okay.

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And then what that is, is they have active motion against full resistance.

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You're kind of giving that resistance and they have active motion and the arm is still up.

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Okay.

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That's some of the ways you guys can assess muscle strength.

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So different types of range of motion.

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So you guys will kind of learn your active range of motion.

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So your active range of motion, the patient can move on their own using their own muscles and their own strength.

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And you have your passive range of motion, which in kind of, it helps, you need help.

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The patient needs help, right?

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They need either assistance from a physical therapist or maybe a device, okay?

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They might need help with that passive range of motion, okay?

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So you guys will kind of realize as you're assessing different types of clients with a weakness, or maybe they've had a stroke or they have residual weakness on the right side.

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It can be very difficult sometimes to assess range of motion because they have a lot of limitations.

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You guys need to be familiar with what type of movement is associated with each joint.

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Not all joints can move in the universal ways that I mentioned.

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Different joints move different ways.

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What is its normal range of motion?

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For patients that have limitations or weakness or pain, be gentle with them and be patient with them and say, do the best you can and we can still collect our data and do an assessment, but do the best you can.

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If you guys are assessing patient's range of motion, so flexion or extension, especially kind of where part of the arm is, where the bend of the arm is, there should be no tenderness, no pain, or something called crepitation.

00:31:29

So crepitation is bone on bone, rubbing,

00:31:35

due to overtime of cartilage being worn out.

00:31:39

Okay.

00:31:39

Or maybe like an air bubble sound.

00:31:41

So have you guys ever tried to like pop your joints or try to pop your hand?

00:31:47

I just did that right now.

00:31:49

It makes like a popping sound.

00:31:51

Okay.

00:31:53

If patients are, you know, doing that range of motion, they should not have any crepidation, which is that bone on bone rubbing.

00:32:00

So you guys might, you know, they might be, it might be painful for them.

00:32:05

Okay.

00:32:07

So I just don't want you guys to confuse crepidation with maybe normal types of cracks that we can hear with our ligaments or tendons during types of motion.

00:32:21

So this is like if I'm sitting in my chair for a long time and I get up and maybe my knees crack a little bit, that doesn't mean I have any joint issues or if I crack my hands, it doesn't mean I have any joint issues.

00:32:33

What we mean is if you are doing that type of assessment.

00:32:39

Okay.

00:32:40

And you know, you are faced with the patient saying, oh my gosh, it's really tender.

00:32:45

It really, really hurts.

00:32:47

Okay.

00:32:49

Or I feel like you're kind of hearing that cracking and they're having pain associated or tenderness with it.

00:32:56

That is an abnormality.

00:32:58

You will have to make sure that that is documented.

00:33:02

Okay.

00:33:04

So your inspection of your posture and gait.

00:33:08

So I guess you see there's different types in that photo that you will see that is good posture or bad posture.

00:33:14

But when you are inspecting your client's posture and gait, you're assessing how straight are they standing up and how smooth and rhythmic are they moving when they walk.

00:33:27

So when we're assessing our posture,

00:33:31

You can do that by observing the client's posture while they're standing.

00:33:35

Their feet should be together.

00:33:37

You're noting is the alignment of the head, the trunk, the pelvis, and the extremities all straight.

00:33:47

And you can also assess your patient's posture by also doing that sitting posture.

00:33:56

So when we're assessing posture, we're assessing are they hunching over, right?

00:34:02

So remember we talked about whether older adults in our previous chapters, sometimes they can kind of hunch over a little bit, right?

00:34:10

We're assessing if there's any abnormalities.

00:34:14

So your lordosis, your scoliosis, your kyphosis, okay?

00:34:21

So think about your abnormalities.

00:34:24

So sometimes you guys inspect posture.

00:34:29

Okay.

00:34:31

So people can have poor posture just from being slumped all the time.

00:34:36

Okay.

00:34:37

So also what you're assessing for with that inspection of your posture, I mentioned your lordosis, your scoliosis and your kyphosis.

00:34:48

Okay, those are different abnormal curvatures of the spine.

00:34:52

So the spines can, you know, the spine on a normal person without any of those issues, lordosis, scoliosis, kyphosis, it's going to be kind of a straight line all the way down, kind of straight midline to the body.

00:35:08

If you see an abnormality like a lordosis or scoliosis, okay,

00:35:15

you will see a spinal curvature where it's abnormal to what the findings are and it can cause posture issues.

00:35:23

Gait.

00:35:25

Gait is how smooth and rhythmic are they moving, right?

00:35:29

It's kind of how we normally walk.

00:35:33

So you can assess if the patient walks in the room.

00:35:37

It should be smooth and equal on both sides.

00:35:42

You will see many types of abnormalities, like with your Parkinson's disease, where they may shuffle when they walk.

00:35:49

It's called a shuffling type of gait.

00:35:51

So they don't walk very smooth and rhythmic.

00:35:53

They just shuffle a little bit.

00:35:56

You guys will learn more types of gait in the upcoming chapters.

00:36:03

I just want you guys to understand that if we're looking for abnormalities in gait, just the basics.

00:36:10

Are they evenly weight bearing?

00:36:12

What that means is, are they evenly bearing weight on both of their feet?

00:36:16

Or are they bearing weight only on one foot and then they're lifting the other foot because they don't want to put any weight on that foot?

00:36:27

You guys can do something called a nudge test.

00:36:33

A nudge test assesses a client's risk for falling.

00:36:37

So what you can do to perform a nudge test is you can stand behind the client and put your arms around the client while you kindly just gently nudge them at the sternum.

00:36:51

Okay.

00:36:51

Normally if you do kind of do this assessment on a client, just tell them what you're doing.

00:36:57

But normally the client does not fall back.

00:37:02

Okay.

00:37:02

And they just maintain their position.

00:37:05

Their strength and maintain that position.

00:37:08

Your abnormal finding is sometimes they can fall back easily and kind of lose their balance.

00:37:14

Can be related to Parkinson's.

00:37:17

Okay.

00:37:19

So that is kind of what you guys are thinking about when you guys are inspecting your posture and gait.

00:37:28

So now this is kind of your inspection and palpation of your temporomandibular joint.

00:37:33

So your temporomandibular joint is kind of the point in your face, kind of in the mouth area, kind of right below, kind of in between the ears, kind of down to the jaws.

00:37:48

That's your temporomandibular joint.

00:37:51

Or we could say TMJ.

00:37:52

So when we're assessing the temporomandibular joint,

00:38:01

What you can do is you can place the tips of your first two fingers in front of the ear and ask the client to open and close their mouth.

00:38:10

Open and close.

00:38:15

Also, what you're assessing with TMJ is moving their jaw from side to side.

00:38:26

Are they able to move it from side to side?

00:38:30

So your temporomandibular joint can move by protracting or retracting, right?

00:38:36

So remember, if we protrude or protract out, that means we can have them extend their jaw outward.

00:38:46

And then retraction is kind of having that jaw go back in.

00:38:55

And then you have your lateral movement of side to side.

00:39:00

You can move it to the right, move it to the left.

00:39:05

And then the movement of your jaw with opening, ah, closing, open and closing.

00:39:14

So sometimes you guys may hear snapping or clicking and that's totally normal and that can be okay.

00:39:23

But what is abnormal?

00:39:26

Remember, our normal is we should have a smooth motion of the mandible, so they should be able to open and close.

00:39:31

They should be able to protract or move it, push it out, or push that jaw out, and then retract and push it back in and kind of move it side to side.

00:39:44

If they're having issues or they have any abnormalities,

00:39:48

So say they have a decreased range of motion, so they might not able to laterally move their jaw side to side.

00:39:54

It's pretty painful for them.

00:39:56

Or there's swelling and tenderness when they are associated with moving the TMJ.

00:40:03

That could be related to arthritis.

00:40:09

So TMJ dysfunction sometimes may cause decreased range of motion.

00:40:16

Sometimes a clicking, popping, or grating sound if you're assessing this type of range of motion.

00:40:24

So now we're talking about our inspection, palpation, and range of motion of our circular, thoracic, and lumbar spine.

00:40:32

So remember, this is what I was talking about with your lordosis, your kyphosis, scoliosis.

00:40:38

How are we inspecting the movement of the spine?

00:40:46

When we're assessing the movement of our spine or the movement of our patient's spine, this is going to be your flexion and hyperextension.

00:40:54

To test the range of motion of our cervical spine, you're going to have the client do a chin to chest method or the flexion.

00:41:05

Flexion is a chin to chest method.

00:41:08

What you're doing is you're going to instruct your client

00:41:12

take your chin and go down to your chest, downward, so that's your flexion.

00:41:19

Hyperextension, I like to say look up at the sky, look up at the ceiling, so you're kind of pushing the head back.

00:41:29

So your flexion is chin to chest, and your hyperextension up to the ceiling.

00:41:39

What you guys are going to find when you're assessing that cervical spine, your normals.

00:41:44

You're going to have a firm, smooth movement.

00:41:47

There's not going to be any tenderness if you kind of move, if you flex or hyperextend.

00:41:56

So your abnormal findings, if a patient has a compression fracture, there could be pain or tenderness because of muscle strain.

00:42:03

So when they try to do this type of range of motion.

00:42:09

Also something called lateral bending.

00:42:13

So lateral bending is you're asking the client to touch each ear to the shoulder on that side.

00:42:26

So what we're doing with lateral bending is what you're doing is having the patient kind of move their neck towards their shoulder to the right.

00:42:40

Okay?

00:42:41

And then move their neck towards their shoulder till their ear touches the left shoulder.

00:42:50

Okay?

00:42:51

So what you're assessing is that range of motion with lateral bending.

00:42:55

So you're going to have the client touch their ear to their shoulder on the right, touch their ear to their shoulder on the left.

00:43:03

That's lateral bending.

00:43:06

Okay.

00:43:07

And then your rotation.

00:43:09

is going to have the client turn right, turn their head to the right, turn their head to the left.

00:43:17

So the range of motion that you guys will see with your cervical, thoracic, and sometime in your lumbar spine is flexion and hyperextension.

00:43:30

So chin to chest, and I'm looking at the ceiling.

00:43:34

Your lateral bending, so ear to

00:43:38

Kind of touching their ear to their right shoulder, touching their ear to the left shoulder.

00:43:45

And then your rotation.

00:43:47

So moving your head to the right, moving your head to the left.

00:43:53

What you guys are assessing in these types of assessments is the spine straight.

00:43:59

Is there any pain?

00:44:00

Is there any tenderness?

00:44:04

So your abnormals that you guys may see for lateral bending.

00:44:07

Is there a limited range of motion with our lateral when we ask our patient to, when we're doing a lateral bending assessment, right?

00:44:19

Is there a limited range of motion?

00:44:21

Is it related to osteoarthritis?

00:44:23

Okay.

00:44:24

Dis degeneration.

00:44:26

Okay.

00:44:28

Rotation.

00:44:29

Do they have injuries in their neck?

00:44:31

Osteoarthritis, kind of the same thing.

00:44:34

Okay.

00:44:35

So when you're inspecting your cervical and lumbar spine, is the spine straight?

00:44:44

Does the spine curve a little bit like you can see in scoliosis?

00:44:49

That's what we're looking for with our thoracic and lumbar spine.

00:44:55

I have kyphosis, lordosis, and scoliosis described here for you guys to review.

00:45:04

if you want to know the differences for those.

00:45:11

So you have different movements of the shoulders when we're thinking about the range of motion.

00:45:16

So you have flexion and extension.

00:45:20

So can they forwardly flex their shoulder up?

00:45:26

And can they extend their shoulder back towards the posterior of their body?

00:45:34

Okay.

00:45:34

And can they hyperextend as well so they can go 50 degrees more?

00:45:39

Okay.

00:45:41

There's abduction and adduction.

00:45:45

Okay, so you guys remember, abduction is away from the body.

00:45:49

So can the shoulder go up away from the body?

00:45:53

And can the shoulder go down towards the body?

00:45:58

So kind of towards touching the hip and touching the legs.

00:46:03

abduction away, adduction towards the body.

00:46:07

Okay, then you guys see you guys also have your internal rotation and external rotation.

00:46:15

So your internal rotation, okay, with the internal rotation, are they able to move towards the body?

00:46:30

And then external rotation, are they able to move outside of the body or kind of away from the body?

00:46:36

That's your internal and external rotation.

00:46:41

What you're assessing, especially with our shoulders, is the head of the humerus and our shoulder, the ball and socket joint, are they able to move?

00:46:54

So sometimes it can be pretty painful.

00:46:58

So if you are assessing these patients and they have limited range of motion, okay, so they have limited range of motion, it could be related to various things, right?

00:47:14

So for example, if you're trying to assess flexion, extension, abduction, abduction, okay,

00:47:25

If they have painful or limited motion, it could be related to atrophy.

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So the muscles are atrophying, there's muscle weakness related with a rotator cuff tear.

00:47:41

Also, when we're inspecting and palpating the shoulders or the arms, your shoulders should be symmetrical and round.

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There should be no bulges, there should be no masses.

00:47:53

No redness, no swelling, no heat.

00:47:56

So if you're feeling that I'm palpating that temperature, it should not be warm.

00:48:00

It should be warm enough for the patient's appropriate temperature, but it shouldn't be hot right in that set because that could be inflammation.

00:48:10

OK.

00:48:12

So tenderness, swelling and heat could be related to shoulder strain or any sort of sprain.

00:48:20

Okay, arthritis, bursitis, or degenerative joint disease.

00:48:25

Just think about when you guys have ever gotten a sprain in your ankle, a sprain in your knee, right?

00:48:32

It gets really inflamed, really swollen, and it's very painful.

00:48:38

Okay.

00:48:40

So you have your elbows.

00:48:43

Your elbows can have a couple of movements.

00:48:45

They can also, your elbows can flex and extend as well.

00:48:52

And you have supination and pronation.

00:48:55

So flexion and extension, what that kind of looks like for your patients, if you're testing range of motion of the elbow,

00:49:05

have their elbow kind of straight out.

00:49:08

So flex towards the body.

00:49:12

Okay, have them flex towards the body and then extend away from the body.

00:49:17

Okay, so you're assessing how well is that elbow joint moving.

00:49:20

Okay, so flex it up towards the body, extend it out towards the body.

00:49:26

Okay, what you're doing is you're inspecting

00:49:31

both elbows in that flexed and extended position.

00:49:34

You're looking for a symmetrical movement.

00:49:38

No structural abnormality, so no swelling, tenderness, pain, heat.

00:49:45

Should be non-tender if you're palpating that area.

00:49:50

If you are assessing range of motion reflection and extension and you feel redness, heat,

00:49:58

or swelling that could be related to arthritis, bursitis as well.

00:50:05

Okay.

00:50:07

Then you have your pronation and supination.

00:50:11

Okay, so pronation.

00:50:15

I like to kind of think of it like moving back and forth.

00:50:20

Okay.

00:50:21

So or downward and upward.

00:50:26

Okay.

00:50:27

So when you're assessing for supination, so supination are they able to kind of face their palm upward?

00:50:34

And then pronation are they able to move their elbow and face their palm downward?

00:50:40

That's supination and pronation.

00:50:45

So all you're doing with supination and pronation is having them hold their arm out, turn the palm down, so for pronation,

00:50:56

Turn the palm back up for supination.

00:51:00

That's what you're doing with your inspection and palpation of your elbows.

00:51:04

Remember, you can flex and extend.

00:51:07

You have supination and pronation.

00:51:11

Now, we're going to talk about the wrists.

00:51:15

So remember, we're going to inspect.

00:51:19

Is the wrist the same size, shape?

00:51:23

Is it symmetrical?

00:51:26

Okay.

00:51:27

You can palpate.

00:51:29

You can inspect any differences in symmetry of the wrist by just having the patient kind of stick the wrist out, right?

00:51:37

See if there is an asymmetry.

00:51:40

Is there any shape?

00:51:41

Is there bulges?

00:51:42

Is there masses?

00:51:45

Is there any sort of trauma?

00:51:47

Okay.

00:51:49

Is it symmetrical?

00:51:51

Kind of mentioned that already.

00:51:52

What does the color look like?

00:51:54

Is it appropriate for ethnicity?

00:52:00

Then you can also palpate the wrist for tenderness or nodules.

00:52:08

So what you're assessing for, what you can do is just lightly, you can lightly touch with two fingers and just lightly feel there's any tenderness or nodules in that area.

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Sometimes if you guys are inspecting and you're palpating, swelling can be seen with rheumatoid arthritis.

00:52:29

There could be tenderness and nodules as well with RA.

00:52:36

There could be signs of wrist fractures if you're assessing the patient and you're palpating and they have pain, tenderness.

00:52:48

The types of motion that the wrist can do is flexion and extension.

00:52:56

Hyperextension, adduction only, and radial and ulnar deviation.

00:53:07

So when we're thinking about flexion and extension of the wrist, what you're doing is asking the client to bend the wrist down,

00:53:19

for flexion and then up straight for extension.

00:53:25

Then what you're going to have the client do is hold their wrist straight up and move the hand outward and inward to assess for radial or ular deviation.

00:53:41

So what you're doing is having them move, hand outward

00:53:48

and N-ward.

00:53:53

Also, your wrists can also do adduction of the wrist as well.

00:54:07

So like I mentioned before, your wrist should be symmetrical without tenderness and swelling.

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You can also perform something called a squeeze test.

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And what you could do is just squeeze the client's knuckle joints.

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So kind of hold your hand around that knuckle and you can squeeze it and they should have no pain with that.

00:54:36

If they have extreme pain, could be related to rheumatoid arthritis or psoriatic arthritis that they can have in the hand.

00:54:48

So carpal tunnel, we're talking about carpal tunnel.

00:54:51

Carpal tunnel is one of those disorders that can be tested by something called the Phalen's test.

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So carpal tunnel can be kind of like a debilitating issue in the wrist.

00:55:06

If someone works in an office and they're typing a lot, they can actually have carpal tunnel where it just feels like a sharp pain kind of going towards their hand.

00:55:17

So to assess for carpal tunnel syndrome, you guys can do the test called the Fallon test.

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What you're going to do is ask the client to place the back of both of their hands, so the dorsal part of their hands, against each other so they're touching each other.

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while flexing the wrist 90 degrees and the fingers are pointed downward and their wrists are dangling.

00:55:41

You're going to have the client hold this position for one minute, and it produces no symptoms in the normal hand.

00:55:54

There should be no tingling, no numbness, no burning or pain in both of the hands.

00:56:01

If your patient says, oh my goodness, I have numbness and tingling on my right hand within a minute, that part or that part or that hand could have that issue with carpal tunnel.

00:56:18

So that is your Fallon test.

00:56:21

So the test will produce numbness and burning in a client in the spot that the carpal tunnel is occurring.

00:56:28

That's your Fallon's test.

00:56:32

The next way you can assess for carpal tunnel is called tunnel sign.

00:56:37

So tunnel sign is what you're doing is you're lightly percussing.

00:56:43

So you're lightly percussing over the median nerve, which is kind of the inner aspect.

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I like to say midline of the wrist.

00:56:53

What you're doing is direct percussion at the wrist.

00:57:01

So when you're doing your tunnel sign, you're just lightly going to kind of tap or percuss that wrist area on the median nerve.

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If it shows up that they have a tunnel sign, or it results in pain, numbness, or some sort of impaired function of the hands and fingers, so like a tingling or shock sensation,

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With that percussion, that is a tunnel sign that there is some carpal tunnel syndrome.

00:57:39

If you're assessing for tunnel sign on a patient that's maybe no carpal tunnel, there's usually no symptoms, so no tingling, no shock feeling at the hands, nothing like that.

00:57:56

Now we're looking at our inspection palpation and how does our range of motion, what can we do if we're assessing range of motion in our hips?

00:58:07

There's flexion, with their knee flexed and their knee extended.

00:58:14

I'll describe all of this in a little bit.

00:58:16

There's extension and hyperextension, there's circumduction, there's internal and external rotation,

00:58:25

and abduction and adduction.

00:58:28

Okay, so first, let's talk about inspection.

00:58:32

Right before we do range of motion, let's talk about inspection.

00:58:36

So first, what we're going to do is when we're kind of looking at the patient's hips, look at them with the client standing.

00:58:43

Okay, inspect the symmetry and the shape.

00:58:46

Okay, is it symmetrical?

00:58:49

Right?

00:58:49

Because we'll hear this lot, is it symmetrical?

00:58:53

Okay.

00:58:55

You're going to assess if the thoracic curvature is straight as well.

00:59:02

You can palpate for stability, so you can kind of palpate the hips.

00:59:06

Is it stable?

00:59:07

Is there tenderness?

00:59:08

Is there any crepitus?

00:59:11

Hips should be stable, non-tender, without crepitus.

00:59:15

Remember that kind of popping, cracking noise.

00:59:21

So usually if your patient is not able to tolerate standing or they have difficulty standing, they have an area where it's a little bit deformed, so it's not like symmetrical for, you know, if you're assessing the right hip is not symmetrical to the left.

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There could be hip fracture, okay?

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If they have any tenderness, they have any edema, so swelling at that site, redness, okay, could be related to hip inflammation, hip inflammation.

00:59:49

which could indicate they might need some sort of hip replacement or degenerative droid disease.

00:59:59

So let's talk about the movement.

01:00:04

So the movement with the hips is flexion with knee flexed.

01:00:11

So hip flexion with knees flexed is the way I like to assess this is have the patient lie supine.

01:00:17

Okay.

01:00:18

I like to have this supine with the patient's head of the bed 30 degrees at least.

01:00:21

I never like to have the lay, never really like to let the bed be completely flat.

01:00:27

Okay.

01:00:30

So hip flexion with knees flexed is you have the patient supine.

01:00:36

Okay.

01:00:38

You're gonna have the patient raise their knee up.

01:00:41

I'm sorry, you're gonna have the patient raise

01:00:44

kind of their hips towards their head and then kind of have their knees kind of bent over.

01:00:54

That's their hip flexion with their knee flexed.

01:00:58

And then the knee extended.

01:01:03

Hip flexion with the knee straight is you're going to have the patient raise their

01:01:10

raise their foot up or raise their leg up towards their body.

01:01:16

They're going to flex with that knee straight up.

01:01:21

It's not bent, it's straight up.

01:01:28

Then you have your extension and hyperextension.

01:01:35

Your extension or hyperextension, you can have

01:01:39

the patient laying prone, so kind of on their belly.

01:01:47

So with your extension, what you're going to do is you're going to have the patient move their legs up.

01:02:02

And then hyperextension, see if they can hyperextend their foot a little bit higher than extension, at least 15 degrees higher.

01:02:14

Then you have your circumduction.

01:02:18

Are they able to move their hips in a circular motion?

01:02:25

And then your internal and external rotation.

01:02:30

I like to have the patient back in a supine position where their back is laying down on the bed.

01:02:38

And I like to have their knees flexed.

01:02:42

And then internally rotate to their body.

01:02:45

So it rotate towards their body and then externally rotate outside of the body.

01:02:55

And then abduction and adduction.

01:02:59

So abduction, I like to have them lay supine.

01:03:03

And then what you're going to have them do is have their knees straight, their hips, and have them move kind of like make a triangle with the body.

01:03:19

With abduction, get their move both legs out till they make a triangle.

01:03:27

And then adduction.

01:03:28

So away from so abduction away from the body, adduction towards the body.

01:03:33

So kind of close up that triangle.

01:03:36

Okay.

01:03:39

So that is your range of motion of your hips.

01:03:43

Now we're going to talk about our knees.

01:03:46

Okay.

01:03:47

So what I like to inspect the knees for our abnormality is when I'm sizing and

01:03:55

or assessing size, shape, symmetry.

01:04:02

Sometimes I like to have the patient laying supine on their back.

01:04:07

I'm assessing if there's any redness, tenderness, or any sort of swelling that I can see.

01:04:16

When I start thinking of palpation, I kind of like to isolate the knees

01:04:26

with two of my fingers, and what I'm assessing is I'm palpating for any tenderness, any warmth, or assessing for nodules.

01:04:38

So I'm seeing if the knees are symmetrical, kind of those kind of assessments.

01:04:47

Should be non-tender, it should feel cool, shouldn't feel warm.

01:04:53

should feel warm, but it shouldn't feel hot.

01:04:54

Like there's swelling, right?

01:05:00

When I'm thinking of the movements, I can do this if the patient can tolerate it.

01:05:08

I can have the patient do this in a standing position, okay?

01:05:14

But most of the times I have done an assessment for flexion extension of the knees, I have done it if the patient is lying supine in bed.

01:05:22

You can do either or.

01:05:25

When you're testing range of motion of the knees, you have flexion and extension.

01:05:30

What you're going to do is instruct a patient to bend their knees toward their buttocks or their back.

01:05:37

That is flexion.

01:05:42

Then you can straighten the knee back out towards

01:05:50

kind of towards the, not the buttocks, but kind of to the front of the body.

01:05:57

And that's going to be your extension.

01:06:00

And then you can also do hyperextension where you can go 15 degrees above.

01:06:06

So your flexion is towards the buttocks or the back.

01:06:12

And then your flexion is just straightening that knee back out straight.

01:06:16

And then hyperextension is going further towards the front of the body.

01:06:22

Usually that's your hyperextension.

01:06:28

Lastly, we're going to go over inspection, palpation, and range of motion of our feet and our ankles.

01:06:37

So you guys can see there is a trend here.

01:06:39

If we're inspecting, palpating, we're feeling for warmth, tenderness, nodules, pain, because that's not normal.

01:06:47

right?

01:06:48

Where that's abnormal for us, okay, when we're doing these assessments.

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So you're gonna inspect.

01:06:57

When we're inspecting the client's feet or ankles, you can do it sitting, standing, you can do it walking, okay, if they can tolerate it.

01:07:05

Always remember, if you're doing any sort of range of motion, just do it until as much as the patient can take, okay, sometimes they're having pain.

01:07:15

they might not be able to perform that range of motion.

01:07:17

Just be very patient.

01:07:21

So usually what we're assessing with this is are the feet and the ankles, are they symmetrical?

01:07:32

Are the toes symmetrical?

01:07:33

Your toes should point forward and it should lie flat.

01:07:39

They shouldn't be smooth, should be free of corns, calluses,

01:07:45

Okay.

01:07:47

Weight bearing should be midline.

01:07:48

So if they're standing, they should have midline weight bearing to where they can weight bear on both right and left feet.

01:07:54

Okay.

01:07:55

Should the toes should be smooth, feet should be smooth.

01:07:59

Okay.

01:08:00

If we're kind of looking at the feet and we're noticing tenderness, swelling around the bony prominences, okay, or crepitus,

01:08:11

Make sure to palpate those and make sure that there's no any worsening issues that we need to assess.

01:08:19

So it should be no pain, heat, swelling, nodules should be noted.

01:08:25

And then the ankles.

01:08:28

The ankles are the most common site for getting sprains.

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That could be related to a torn ligament.

01:08:39

which are connecting bones in a joint.

01:08:43

I will tell you guys, I had a sprained ankle when I went to Florida and I sprained it pretty badly and I couldn't walk on it for two days.

01:08:52

So it is really painful if there are sprained ankles.

01:08:58

So some symptoms, signs that be accommodated with that.

01:09:03

So it could be tender, tender with movement.

01:09:08

So say you're trying to do like rotation, kind of that rotation of the foot.

01:09:14

Some patients can't do that because it hurts that bad, right?

01:09:17

Is it tender?

01:09:18

Is it painful?

01:09:19

Is it red?

01:09:20

Is there a swelling in the metatarsal metaphalangeal joint with gouty arthritis?

01:09:30

So your ankles, is there any swelling if they try to rotate that or move that?

01:09:37

Is there any swelling kind of in that part of the toe, the big part of the toe that can be seen with gout?

01:09:44

Okay, so movement.

01:09:47

There is something called plantar flexion.

01:09:51

Okay, plantar flexion and dorsiflexion.

01:09:57

So plantar flexion, you guys will assess plantar flexion and dorsiflexion and head to toe.

01:10:05

This is also a type of range of motion that I have done when I have assessed patients who have had stroke.

01:10:13

So plantar flexion and dorsiflexion.

01:10:18

So for dorsiflexion, you're going to have your clients point the toes upward towards the sky.

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And then downward is your plantar flexion.

01:10:31

So like to say, with plantar dorsiflexion, go toes to your nose.

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And then if you're going plantar flexion, you can go press like you're pressing on a gas pedal.

01:10:49

The movements of your feet.

01:10:52

You have inversion and eversion.

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So what you're doing with eversion is you're turning the soles outward, away from the body.

01:11:09

And then what you're doing with inversion is you're turning it inward towards the body.

01:11:24

So that is the movement that you can have with your feet and your ankles.

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An addition here that is not added is abduction, then inward, abduction and adduction.

01:11:45

So your foot can rotate outward.

01:11:47

So that's your abduction.

01:11:53

And then your foot can rotate inward, which is your adduction.

01:12:03

Alright, we're getting to our older adult considerations.

01:12:09

So I mentioned this before in our previous slides when we're considering activities of daily living, right?

01:12:18

What are older adults a greatest risk of is falls, right?

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Remember I talked about that bone density scan, we can get

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That assesses how firm, how hard, and how much of our bone is still not softened, things like that.

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Our bones can lose density over time with age, which increases the risk for fractures.

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Bones can break.

01:12:53

Bones can also start demineralizing like an osteoarthritis.

01:13:01

So it could be worsening in their wrists, their hips, and vertebrae.

01:13:06

So older adults could be more prone to maybe getting kyphosis, lordosis, scoliosis.

01:13:18

So remember your kyphosis.

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It could be that rounded convex of the thoracic curvature.

01:13:28

your scoliosis or could be a lateral curvature of the spine.

01:13:40

So think about that.

01:13:42

The most common thing that you guys will see in surgical floors are patients or clients that are 80 plus and they have had to get hip replacements.

01:13:58

It is because they have their bone density is decreased.

01:14:03

They say, for example, how to fall at the skilled nursing facility and they fell on their hip.

01:14:08

Now that hip is fractured, they have to get a replacement.

01:14:14

So women, men have osteoporosis, arthritis are at greatest risk for fractures.

01:14:21

Remember, older adults are a little bit slower.

01:14:23

They have that decreased flexibility.

01:14:27

A lot of them might have an impaired sense of positioning, so they might not know how to position correctly and have a fall.

01:14:40

They might have a bow-legged appearance.

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A bow-legged appearance is when it's related to your osteoarthritis and that cartilage is worn down.

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And that sometimes they can kind of look bow-legged in the area of their joint.

01:15:00

So those are your older adult considerations.

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So what is a functional assessment?

01:15:10

A functional assessment is an assessment that physical therapists can do, nurses can do, or your physicians to ensure that they are safe to go home.

01:15:24

For example, say a patient comes in.

01:15:29

Patient had a fall on their hip.

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They had a hip replacement on the right side.

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What are the things we are assessing functionally for them to be safe at home so that fall doesn't happen again?

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A couple of things for those patients that have

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advanced changes with arthritis, musculoskeletal weakness, disabilities.

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We need to perform a functional assessment of activities of daily living.

01:16:04

So these could be fine movements.

01:16:09

These could be precise movements.

01:16:12

These could be gross movements.

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So what I mean by fine movements, are they able to pick up a spoon or fork?

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Are they able to eat food?

01:16:23

Are they able to pick up their fork to eat food?

01:16:28

Those are your fine movements, very small fine movements.

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It could be asking the patient, can you button up your shirt?

01:16:38

Those are fine movements.

01:16:41

And you have your gross movements.

01:16:42

That's like your movements where you're getting up and actually walking to the bathroom.

01:16:48

Right.

01:16:49

You're helping, you're, you're helping these patients when you do a functional assessment.

01:16:55

So you can see what, what are the things, what can they not do?

01:16:59

What did they need help with?

01:17:01

Okay.

01:17:02

So we're applying our range of motion exercises to see what are the limitations.

01:17:08

Okay.

01:17:08

What they, what can they do?

01:17:09

What can't they do?

01:17:11

Okay.

01:17:12

Are they able to come back home independently?

01:17:18

to take care of themselves or do they need rehab or help from someone to come to their home to help them out for the first six weeks?

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Do they have family members that could help take care of them?

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These are the things that we're doing when we're assessing a functional assessment.

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So you guys can see the goal is to determine is it adequate and safe

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for these patients to come back home independently like they used to.

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I will tell you guys, falls, fractures, patients that have had knee replacements, hip replacements.

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And if it's severe enough, their recovery sometimes can take a long time.

01:18:02

And I will tell you sometimes they are not able to be functionally independent at home for a while.

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So they might have to come back or get discharged to a skilled nursing facility with the rehab.

01:18:13

where they need help with movement and kind of honing in those skills that they used to have before the fall and before the surgery.

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And then they can come back home and it can just be like an interim until they need to be back home.

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So those are just some of the things that we can do if we're doing a functional assessment.

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And for some of you that are, you know,

01:18:40

working in clinical and you see surgical floors, you guys will see physical therapists actually do these functional assessments, especially with like eating and drinking and moving and things like that.

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And sometimes that's really cool to see if you're curious about kind of seeing what we kind of do for this assessment.

01:19:00

Just a couple things that we'll talk about in the next couple slides.

01:19:04

Okay, a couple inflammatory conditions.

01:19:07

your rheumatoid arthritis, your ankylosing spondylitis, and your degenerative conditions like your osteoarthritis or degenerative joint disease, and your osteoporosis.

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And each of these we'll talk about in detail in the slides coming up.

01:19:26

So rheumatoid arthritis is one of the first inflammatory disorders we'll be talking about.

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So rheumatoid arthritis

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is a chronic autoimmune disease and it progresses over time.

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So there's chronic swelling and thickening of the metacarpophalangeal and proximal interpharyngeal joints.

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So those are the small joints you have in the fingers.

01:19:55

And what causes what kind of is a classic sign that someone is kind of going

01:20:03

through the process of RA and it's been chronic, it's been happening for a while, is they will have this swan neck deformity.

01:20:13

You guys can see there that is what swan neck deformity looks like.

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That is a classic sign of RA.

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There's that ulnar side and there's different types of, they have limited range of motion on their fingers.

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Okay, this is why we talk about like the fine gross movements, right?

01:20:33

And they also have different types of finger deviations on the ulnar side that you can see with chronic RA.

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So you guys can kind of see that in that photo down there.

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RA, over time, acute rheumatoid arthritis, when it can first start, could develop with tender, painful, swollen, stiff joints, okay?

01:20:57

Hands might be more reddened, more painful.

01:21:03

And signs can vary.

01:21:05

I like to say signs and symptoms can vary.

01:21:10

Over time, if the inflammatory process worsens, it can affect different parts of the body.

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It can worsen over time, like I mentioned before.

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Patients can experience muscle weakness.

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They might experience fatigue, feeling more tired or depressed.

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They might have weight loss.

01:21:36

All these patients can have an elevated temperature.

01:21:43

Sometimes they have lymphadenopathy or swollen lymph nodes.

01:21:47

Okay.

01:21:48

And sometimes those patients that kind of, you kind of assess their fingers, they can have like significant stiffness in their body, in their fingers the morning, in the very early morning, and that can persist for sometimes an hour.

01:22:04

So sometimes you will see these patients take a warm shower in the morning to help relieve some of that pain.

01:22:14

Okay.

01:22:14

The destruction of cartilage and bone.

01:22:17

causes in that inflammatory process really affects not just our fingers, but it can affect our whole body.

01:22:30

A couple things to educate the client on with RA.

01:22:35

Using the right body mechanics and symptom management to help them have decreased pain.

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Maybe don't do heavy lifting.

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Walking to help get more flexibility in those joints.

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So you can see in these photos, and that is pretty chronic RA with those swan neck deformities, ulnar deviation of the metacarpophalangeal joints.

01:23:06

It's hard to pick things up or maybe be able to eat.

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So you might have to teach them what are ways to help decrease that pain.

01:23:15

So that's your rheumatoid arthritis.

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Now, this is your ankylosing spondylitis.

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Ankylosing spondylitis is another one of those inflammatory processes.

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Again, this is chronic.

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So your ankylosing spondylitis could be related to worsening kyphosis.

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It also can be worsened with patients who are males and it could be from birth.

01:23:58

So ankylosing spondylitis is an inflammatory arthritis of the spine, hips and shoulders where inflammation, there's inflammation and tendency for fibrosis and calcification.

01:24:15

At that side of the muscle insertion.

01:24:20

When you're assessing your patient that has ankylosing spondylitis, your patients will have kyphosis or extension of the neck, so they might be hunched over.

01:24:34

Their eyes are more inflamed or uvitis.

01:24:42

Also, their posture might be not

01:24:45

straight.

01:24:46

Their posture is not going to be as straight.

01:24:51

They have less mobility of their spine, so they might have something called anthesis.

01:24:57

They have inflammation and fibrosis at different sites of their muscle.

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It could be at their shoulders, their hips, their knees, or their heels.

01:25:12

Think about your ankylosing spondylitis.

01:25:15

your the patient's actual spine can be fused.

01:25:20

So it's fused.

01:25:24

Okay.

01:25:25

I like to say those fusion of bones.

01:25:27

So instead of having that separation of disc right and the vertebra, it's kind of all fusing together.

01:25:39

Okay.

01:25:39

So the discs are fusing with the joints and the ligaments.

01:25:42

That's your ankylosing spondylitis.

01:25:47

So osteoarthritis.

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Osteoarthritis is the most common form of arthritis and it's kind of related to a degenerative joint disease.

01:26:00

It is just a wear and tear of the joints over time.

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It can affect the joint, the bone, the cartilage, ligaments, fats.

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and that synovium or that lining of the tissues.

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And what is occurring in osteoarthritis is there's degrading of cartilage, and it's progressive, which can change the shape of bones and also cause inflammation, resulting in pain, stiffness, and decreased mobility.

01:26:43

Osteoarthritis can affect a lot of joints, but it's typically associated with your hands, the knees, the hips, also the lower back and neck.

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You guys can see osteoarthritis here in this hand.

01:27:03

There's also kind of those changes as well in the interphalangeal joints.

01:27:08

We call them hebridinodes.

01:27:12

And it's usually a bony overgrowth that can cause that due to osteoarthritis.

01:27:20

Those hybrid nodes can be painless for some people.

01:27:26

So that's one of the signs in OA, especially in the hands.

01:27:34

When we're thinking about causes of osteoarthritis,

01:27:41

We have to think about the risk factors.

01:27:44

Generally, women can get this more, especially over age 50.

01:27:54

Any sort of joint injury, so a bone fracture or a cartilage or tear in the ligament can lead to osteoarthritis.

01:28:05

Using overuse, so you're using the same joints over and over again.

01:28:10

in a job or a sport.

01:28:13

Obesity, so increased weight can also be related to osteoarthritis.

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Think about all the weight in the body.

01:28:24

Excess weight that's adding stress and pressure on that joint.

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And fat cells can promote inflammation, so it's causing more inflammation in the body as well.

01:28:36

Musculoskeletal abnormalities.

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So is there any misalignment of bone structures that can develop, contribute to faster development of OA?

01:28:48

Any weak muscles, genetics, so family history.

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Women are at increased risk for getting osteoarthritis greater than men.

01:28:59

And environmental factors.

01:29:01

So what are those?

01:29:02

What is someone's job?

01:29:03

Do they have a very stagnant desk job where they don't move a lot?

01:29:06

Do they exercise?

01:29:10

Okay.

01:29:11

Are they obese?

01:29:12

What is their bone density?

01:29:14

What type of foods do they eat?

01:29:17

Okay.

01:29:18

Think about these things, especially those factors causing that risk for osteoarthritis.

01:29:25

Okay.

01:29:28

So sometimes again, it's related to stiffness, swelling, okay, and pain.

01:29:37

So osteoporosis.

01:29:41

So osteoporosis is different than osteoarthritis, right?

01:29:49

So what's happening in osteoporosis is the bones demineralize and can become very porous.

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So you guys can see in that picture right there with osteoporosis, it's more porous than that healthy bone.

01:30:05

So that bone is becoming more fragile

01:30:07

and there's less bone density.

01:30:12

The bone loss can occur silently and progressively over time.

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And generally, there's no symptoms until the first fracture occurs.

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So think about how hard bone is, and it's hard to break that.

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But if someone has osteoporosis, fractures and breakage of bones is very common.

01:30:33

It can be very painful.

01:30:39

So osteoporosis can occur at any age.

01:30:44

Normally, that can peak.

01:30:47

Bone mass peaks usually between 25 and 30 years of age.

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So when we're thinking about osteoporosis,

01:30:59

We have to think about what are the things, what is the goal that we have for osteoporosis when we're thinking about promotion and disease prevention?

01:31:10

So screening.

01:31:12

Screening might be super helpful for these patients.

01:31:18

So bone density scans.

01:31:21

Are we treating the appropriate older adults after a fracture?

01:31:30

Osteoporosis is more common in women who are Hispanic, White, or Asian.

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However, it's lower risk for Black and Hispanic individuals.

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They're at lower risk, but they still are at significant risk.

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So risk assessment, a couple risk assessment for your osteoporosis patients.

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Age 65 and older.

01:32:02

I mentioned this assigned female.

01:32:07

You guys see a pattern with osteoporosis, osteoarthritis.

01:32:10

It's mostly common in females.

01:32:13

Risk for falls.

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So if they had a fall, they could be at risk.

01:32:19

Rheumatoid arthritis, hyperthyroidism,

01:32:23

Low bone mineral density.

01:32:25

So low density and their bones that the porous type of bone.

01:32:31

Okay.

01:32:33

Early menopause.

01:32:34

So women before age 45.

01:32:38

Okay.

01:32:39

That's your risk factors that are not modifiable.

01:32:44

Okay.

01:32:44

Your modifiable risk factors.

01:32:46

What are the things we're educating these clients on?

01:32:49

Maybe limiting alcohol.

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Nicotine use.

01:32:54

Don't smoke.

01:32:57

Making sure they have a very good exercise routine.

01:33:02

Keeping their BMIs low.

01:33:08

Make sure they're eating enough calcium and protein in their diet to build up that to decrease the worsening of bone density.

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Patients that are vitamin D deficient, so not much calcium.

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Guys can see sometimes when we think about races or ethnicities, Asians are more prone to being lactose intolerant, so they're not going to have as much vitamin D in their diet because of lactose intolerance, so they have to get that somewhere else.

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So kind of think about the education, ensuring that they're eating enough calcium, protein.

01:33:57

Making sure that they have enough vitamin D in their diet.

01:33:59

So that's your osteoporosis.

01:34:04

So you guys can kind of review this.

01:34:06

I mentioned this in the previous slide, but again, just reiterate screening.

01:34:14

65 and older, make sure we're assessing bone density.

01:34:18

We're making sure we're educating these patients about their risk factors.

01:34:24

What are the things we can do to educate these clients?

01:34:30

Some more education pieces.

01:34:32

I know I kind of mentioned it in the previous slides, right?

01:34:36

But ensuring that these clients get an adequate diet of calcium intake.

01:34:41

So calcium intake that requires maybe milk.

01:34:46

right?

01:34:47

What are the types of foods that are high in calcium?

01:34:53

I'm avoiding being undernutrition.

01:34:54

So maybe a client that has an eating disorder is a higher risk for getting osteoporosis, right?

01:35:02

They're not getting enough protein intake.

01:35:07

Participating regular weight bearing activities.

01:35:09

So that could be walking, just walking around the neighborhood that could be doing very light

01:35:16

weight training, maybe two pound little bells.

01:35:20

Okay.

01:35:21

Avoiding heavy drinking.

01:35:23

So casual drinking is okay once every couple months, but maybe don't educate these clients.

01:35:29

Don't drink so much.

01:35:32

Right.

01:35:35

Making sure that we're also avoiding active and secondhand smoking.

01:35:41

Okay.

01:35:42

So these are some of the things that we can educate our clients on, especially if we're trying to reach the goal of Healthy People 2030 with osteoporosis.

01:35:54

And I like to say when I'm educating these clients, I like to do a risk assessment.

01:35:57

So I like to look at who is at higher risk and how much education do I need versus another person.

01:36:10

So I might look at the risk factor and say, oh, they're overweight and their lactose are not getting enough protein, or they're not, sorry, they're not getting enough vitamin D.

01:36:23

So what are vitamin D rich foods?

01:36:28

So educate those clients on vitamin D rich foods that they can eat to supplement that they can have dairy, okay?

01:36:38

Those foods can include salmon, eggs, mushrooms.

01:36:45

So educate them on their diet, maybe increasing more of those to make sure they can get that intake of vitamin D in.

01:36:53

Every time I educate clients on any sort of disorder or anything like that, I always assess what is their risk

01:37:03

and what will they benefit from my education and I tend to look at it before I start educating because some people already are aware but some are not good stewards of health so you might need more education.

01:37:18

Here's a review question for you guys.

01:37:22

A client is or a nurse is caring for a client who reports pain with internal rotation of her right shoulder.

01:37:34

This discomfort can affect a client's ability to perform which of the following activities?

01:37:41

Mopping her floors, brushing the back of her hair, fastening her bra behind her back, or reaching a cabinet above the sink.

01:37:54

I want you guys to kind of think about that review question.

01:38:02

I kind of think about that.